

Module 14: Macroevolution — Keys to Success

Introduction and Big Picture

While Module 13 focused on changes within a single population (microevolution), this module zooms out completely. **Macroevolution** deals with the big picture: the origin of new taxonomic groups, explosive diversity, and most fundamentally, the creation of brand new species. The core engine behind speciation is reproductive isolation—a mechanism that stops gene flow.

Key Information and Concepts

1. The Biological Species Concept

According to the Biological Species Concept, a **species** is a group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring, but do not produce viable, fertile offspring with members of other such groups. Importantly, if two groups cannot exchange genes successfully, their evolutionary paths are severed, and they become separate species.

2. Reproductive Barriers

Speciation requires genetic isolation. Barriers that prevent different species from successfully mating are categorized based on whether they happen before or after the formation of a zygote (a fertilized egg).

Prezygotic Barriers (Before Fertilization)

These barriers prevent mating attempts outright, or prevent fertilization if mating is attempted:

- **Habitat (Ecological) Isolation:** Two species occupy different habitats within the same area and rarely meet (e.g., one snake lives in water, a closely related one lives on land).
- **Temporal Isolation:** Species breed during different times of day, different seasons, or different years.
- **Behavioral Isolation:** Courtship rituals or behaviors unique to a species act as mating barriers (e.g., unique bird songs or firefly flash patterns).
- **Mechanical Isolation:** Morphological

differences prevent successful mating (e.g., snail shells that spiral in opposite directions so reproductive organs do not align). - **Gametic Isolation:** Sperm of one species may not be able to fertilize the eggs of another species (common in marine animals that broadcast spawn).

Postzygotic Barriers (After Fertilization)

If a sperm does fertilize an egg of a different species, these barriers prevent the hybrid from developing into a viable, fertile adult: - **Reduced Hybrid Viability:** The genes of different parent species interact in ways that impair the hybrid's development or survival in its environment (the offspring dies young or in utero). - **Reduced Hybrid Fertility:** Even if the hybrid is vigorous and survives, it is completely sterile (e.g., a mule, which is the sterile offspring of a horse and a donkey). - **Hybrid Breakdown:** Some first-generation hybrids are viable and fertile, but when they mate with one another (or with either parent species), offspring of the *next* generation are feeble or sterile.

3. Modes of Speciation

Speciation can occur in two primary geographical ways: - **Allopatric Speciation ("Other Country"):** Gene flow is interrupted when a population is divided into geographically isolated subpopulations (e.g., a river changes course, or a mountain range rises, splitting a population in half). Over time, the isolated groups evolve independently through mutation, drift, and selection until they can no longer interbreed. - **Sympatric Speciation ("Same Country"):** Speciation occurs in populations that live in the exact same geographic area. This requires the emergence of a reproductive barrier that isolates a subset of the population without geographic separation. This usually happens rapidly via: - **Polyploidy:** Errors during cell division that result in extra sets of chromosomes. This is extremely common in plants and can create a matching, isolated set of individuals in a single generation. - Habitat differentiation and strong sexual selection can also drive sympatric speciation in animals.

Strategic Tips for Studying

- **Link Micro to Macro:** Remember that macroevolution isn't fundamentally a different process than microevolution. Given enough time, isolated instances of microevolutionary change (drift, mutations, selection) accumulate until a reproductive barrier forms. That threshold is speciation.

- **Categorize Barriers:** Practice categorizing real-world examples specifically as either prezygotic or postzygotic. If sperm hasn't met egg, it is prezygotic.